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WORK INSTRUCTION FOR REVISED TRP-011 (PS-1W) OF
OFFSHORE TACTICAL RESPONSE PROCEDURE

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FINAL APPROVAL

OIH

	WORK INSTRUCTION FOR REVISED TRP-011 (PS-1W) OF OFFSHORE TACTICAL RESPONSE PROCEDURE DOC NO: IS-HSE-WKI-011 REV. 00
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1.0 PURPOSE

This Work Instruction is related to revised ERT Tactical Response Plan TRP-011 Plan for PS1-W, Reference IS Offshore Tactical Response Procedure (IS-HSE-PRC-003). This Work Instruction was prepared and approved based on the Installation of new paint container at PS-1W.

Note: This Work Instruction replaces TRP 011 of IS Offshore Tactical Response Procedure (IS-HSE-PRC-003)

2.0 SCOPE AND APPLICABILITY

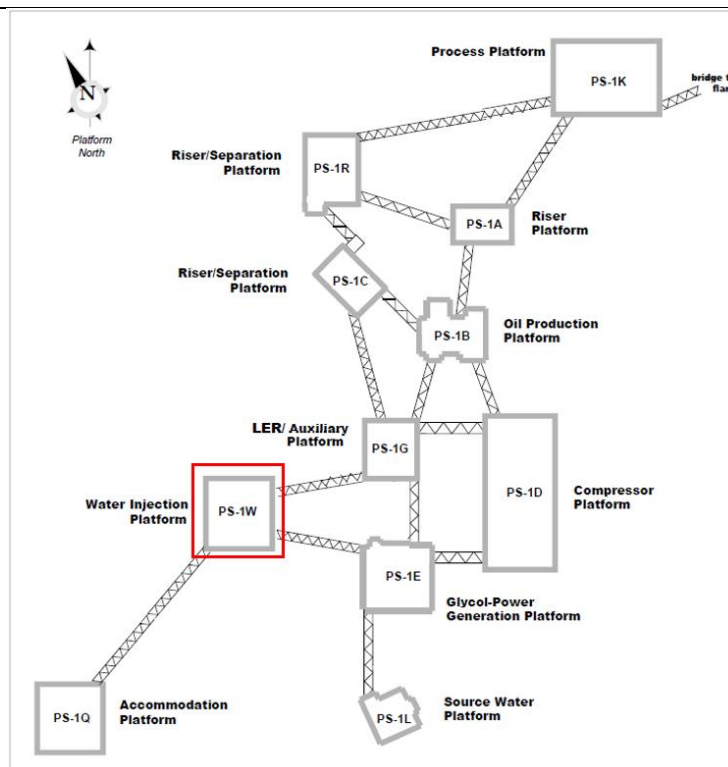
This Work Instruction supports the On-scene Commander and Emergency Response Team's decision-making process and is applicable to the Offshore Installation Superintendent (OIS), their deputy, Central Control Room (CCR) staff and those with the responsibility for offshore site emergency response.

ERT Tactical Response Plan TRP-011

UNIT: PS-1 W

Issued:	
Review Due:	
Version:	

July 2021
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0.2



Hazards

Fire Hazards

- Fuel Gas (Typhoon)
- Paints stored in the container on PS-1W cellar deck laydown area

Hazardous Substances/Materials
- Scale Inhibitor: Corrosive to the skin. Causes burns / Corrosive to eyes. Causes burns.

Special Hazards

- Asphyxiating atmosphere likely to be present, BA sets required for entry

PPE REQUIREMENTS

- Firefighting – Fire Kit, SCBA
- Chemical Spill Response PPE
 - Type C – inc chemical gloves for small liquid spills/releases
 - Type B – for larger spills of Biocide or Scale Inhibitor

Communications

- Tel: 40631 / 40638 / 40697 / 40652 / 40695 (Mechs Office)
- Tel: 40654 (Inst Lead)
- Tel: 40606 / 40690 (E&I Workshop)
- Tel: 40698 (Storekeeper)
- Tel: 40602 / 40694 (LER)
- Tel: 40605 (Lab)

Process/Equipment Information / Major Inventories

ID	Inventory	Pressure	Fixed System	Blow Down
Typhoon Water Injection Pump (GT-4156)	<i>Fuel Gas</i>	<i>90-95 bar</i>	<i>CO2</i>	<i>N/A</i>
Seawater booster pump (P-4155)	<i>Sea Water</i>	<i>9-10 bar</i>	<i>N/A</i>	<i>N/A</i>
Paint container on PS-1W cellar deck laydown area	<i>Paints</i>	<i>N/A</i>	<i>Smoke Detector/Alarm</i>	

Response Guidelines

Worst Case Scenario: Fuel gas leak inside Typhoon Enclosure (potential fire / explosion)

- Initial response is to rely on the effects of shutdown/blowdown, coupled with effects of deluge systems.
- In conjunction with OIS confirm safe to proceed, prior to mobilizing.
- Cool surrounding exposures using onsite water spray fixed systems (deluge) / monitors / fire hoses / branches.
- Confirm CO2 fire system inside the enclosure has activated in the event of fire.
- Monitor internal temperatures on the Typhoon Compressor prior to making entry.
- Use water streams/deluge for gas dispersion and/or suppression.

GAS RELEASE (reference to PIP-05)

- **Select FCP based on wind direction**
- Identify Access / Egress Points prior to mobilizing
- Establish & report gas level readings on approach
- Conduct a Dynamic Risk Assessment of the scene
- Confirm Electrical Isolations are completed
- Use water spray/deluge for gas suppression
- Consider gas dispersion tactics using fixed monitors/hose streams.
- Use water spray/deluge for gas suppression

RESCUE:

- Monitor gas levels
- Confirm status of casualties / missing persons

First Aid

- Wash skin in safety shower for 15mins
- Flush eyes in eyewash station 20mins

FIRE: (reference to PIP-04)

- Confirm activation of fixed suppression system (Typhoon Compressor)
- Cool and protect exposed structures (adjacent and above), Tanks & Vessels using portable hose/branches.
- Monitor Status of Fire Water Pump

HAZMAT SPILL/RELEASE:

- Fire Kit & SCBA/facemask with filter cartridge
- Actions for small spills (<210 litres):**
- Establish zones; Hot/Warm/Cold
 - Establish Emergency Decontamination
 - Identify primary decontamination methods
 - Level B PPE may be required for spills of Scale Inhibitor or Biocide.

Logistics

ERT x 7 – 4 x Team Members (Entry teams), 1 x BA Controller, 1 x Emergency Response Team Leader,
(Plus one BA Technician)

Forward Command Post/Access/Egress

PS10 Bridge to W

Hydrants / Fire Extinguishers

- 2 on Main and Upper Deck
- Main Deck: Portable Fire extinguishers available inside the container (5kg) and outside the container (50 kg DCP type)

Showers

- 1 x Lower Deck
- 1 x Main Deck (near W-G Bridge)

Hazmats

- **Spill Kit** located on Main Deck
- **PPE** - Hazmat Locker on Main Deck
- **MSDS @** Available at chemical storage

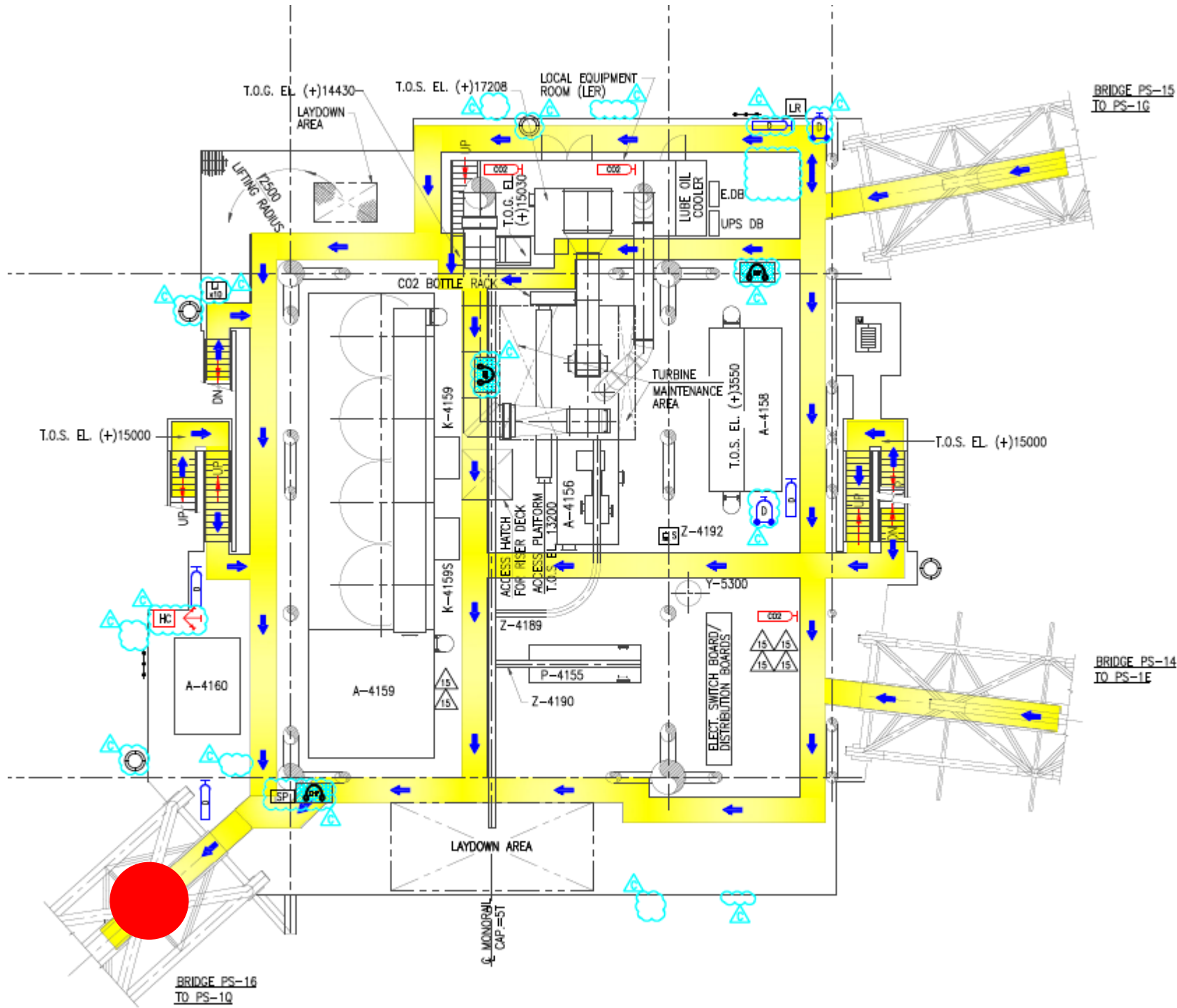
Emergency evacuation

- Life rafts @ 1 x life rafts (South Side)
 - Knotted ropes @ 2 x knotted ropes
 - Boat landing accessible
- Stretcher / First Aid Kit located inside LER

Fire Protection

- Deluge System
- CO2 (Typhoon compressor)

PS-1 W (“Main Deck”) Forward Command Points and Access / Egress Routes



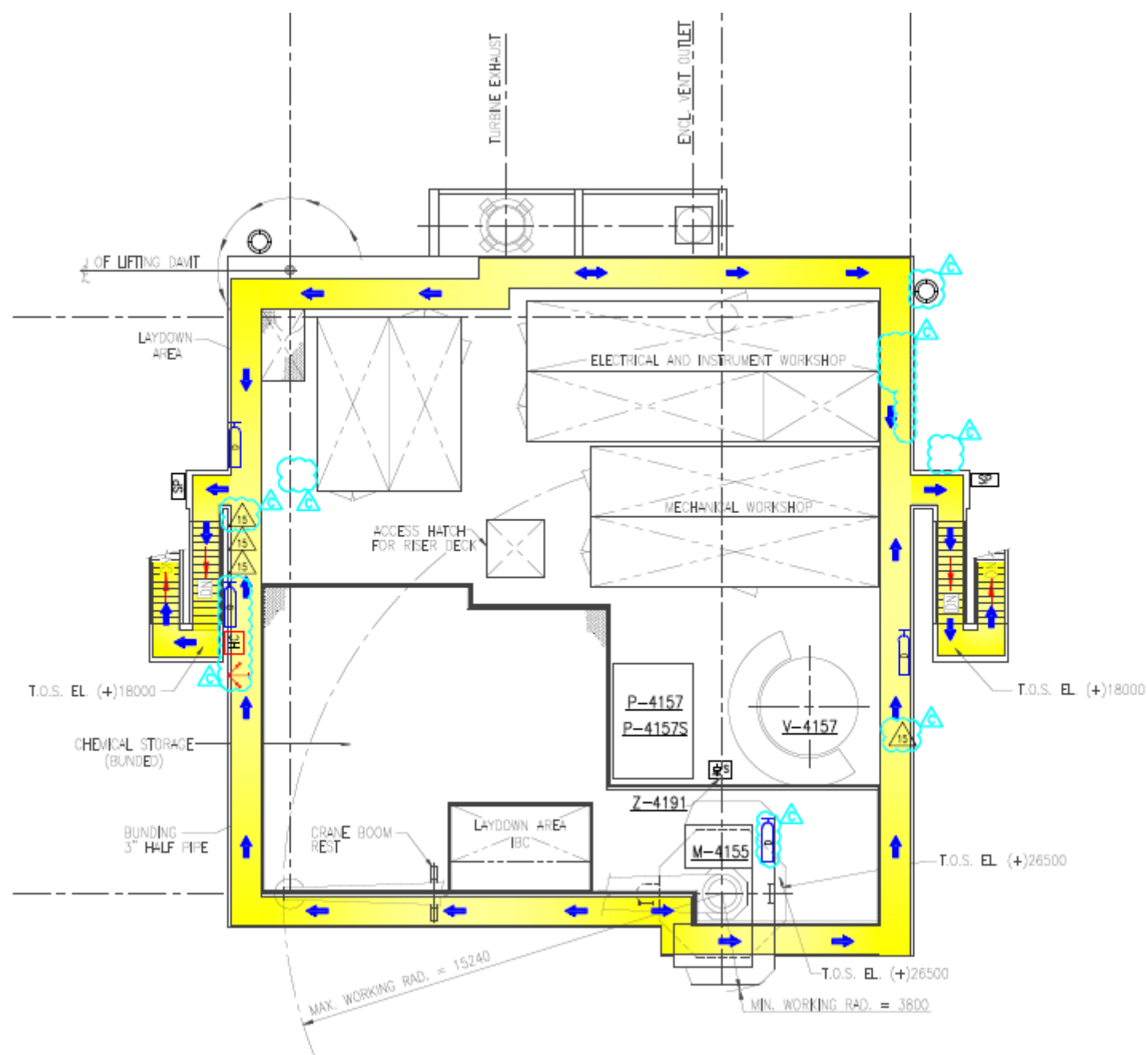
PS-1 W (“Upper Deck”) Forward Command Points and Access / Egress Routes



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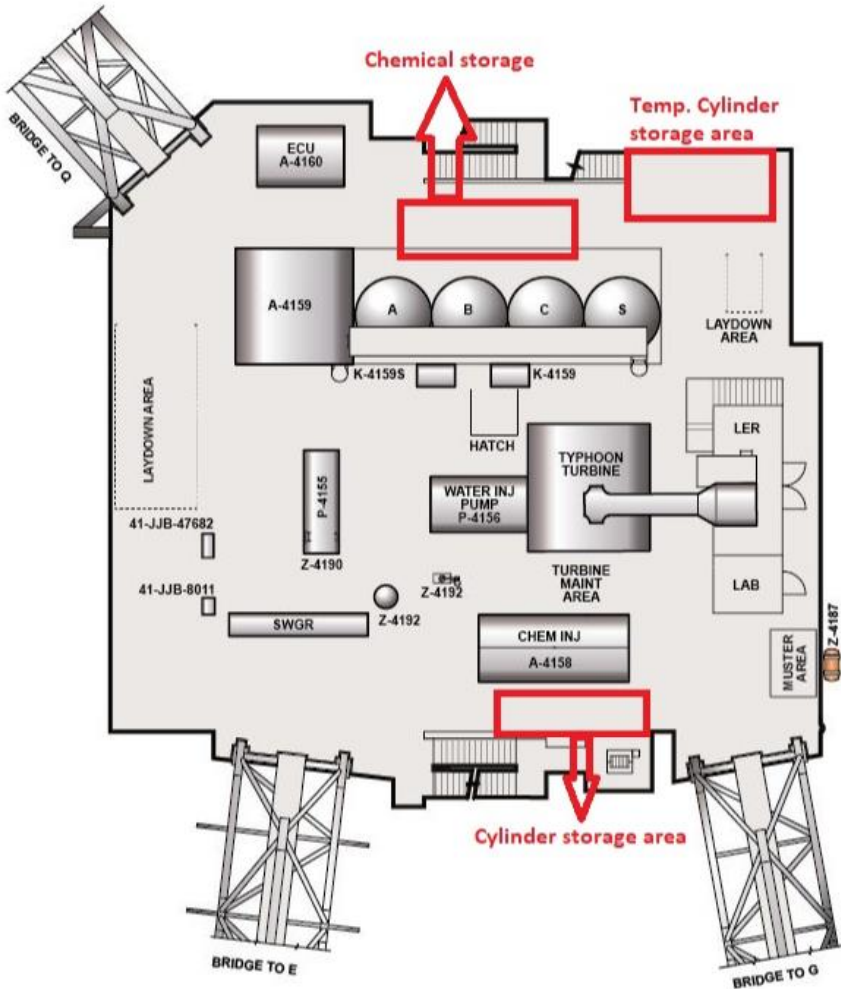
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PS1-W Chemical Spill Response

Chemical Information																
	Chemical Name	Container	Volume	Location	Use											
1	Coagulant	T-4158A	680 ltr	Main deck / Upper deck	Water Injection											
2	Polyelectrolyte	T-4158B	1397 ltr	Main deck / Upper deck	Water Injection											
3	BIOCIDE-G	T-4158C	2196 ltr	Main deck / Upper deck	Water Injection											
4	Scale Inhibitor	T-4158D	6475 ltr	Main deck / Upper deck	Water Injection											
5	Antifoam	T-4158E	2370 ltr	Main deck / Upper deck	Water Injection											
6	Oxygen Scavenger	T-4158F	2196 ltr	Main deck / Upper deck	Water Injection											
7	BIOCIDE-QA	T-4158G	2208 ltr	Main deck / Upper deck	Water Injection											
8	Corrosion Inhibitor	T-4158H	4450 ltr	Main deck / Upper deck	Water Injection											
Actions for small chemical spills (<210 litres): - Establish zones; Hot/Warm/Cold - Establish Emergency Decontamination - Location of Safety Showers & Eyewash stations - Identify primary decontamination methods - PPE is dependent upon the size of the spill and the probability of contamination. - Mitigate the spill; - o Stop leak if possible & Contain - o Confine - o Divert - o Absorb - Absorb with dry material/absorbent pads or socks Place contaminated pads in a suitable waste container																
Decontamination - Wash gloves/boots with copious amounts of water.																
Chemical Spill Clean Up Measures																
	Chemical Name	Hazardous Classification	PPE Class SMALL SPILL	PPE Class LARGE SPILL	Use spark proof and explosion proof tools	Do not allow to dry out, flush thoroughly with water	Dilute with water and mop up	Neutralize (refer to SDS)	Collect spill mechanically by pump or vacuum	Absorb with an inert dry material	Evolving gases spread at ground level	Suitable extinguishing media	Location	Quantity		
	Coagulant	Irritant	C	C								Dry chemical	Main deck / Upper deck	680 ltr		
	Polyelectrolyte	Irritant	C	C								Water spray	Main deck / Upper deck	1397 ltr		
	BIOCIDE-G	Irritant	B	B								Water spray	Main deck / Upper deck	2196 ltr		
	Scale Inhibitor	Corrosive/Toxic	C	B								Water spray	Main deck / Upper deck	6475 ltr		
Chemical Spill PPE Matrix										Suggested Zoning of the Spill Response Area						



- There are two levels of PPE available on PS-1 W.
- Level B – Liquid protective suit (Spill kit located on PS1W)
 - Level C – light-weight protective suit in Spill Kits. Chemical protective gloves in cabinets.

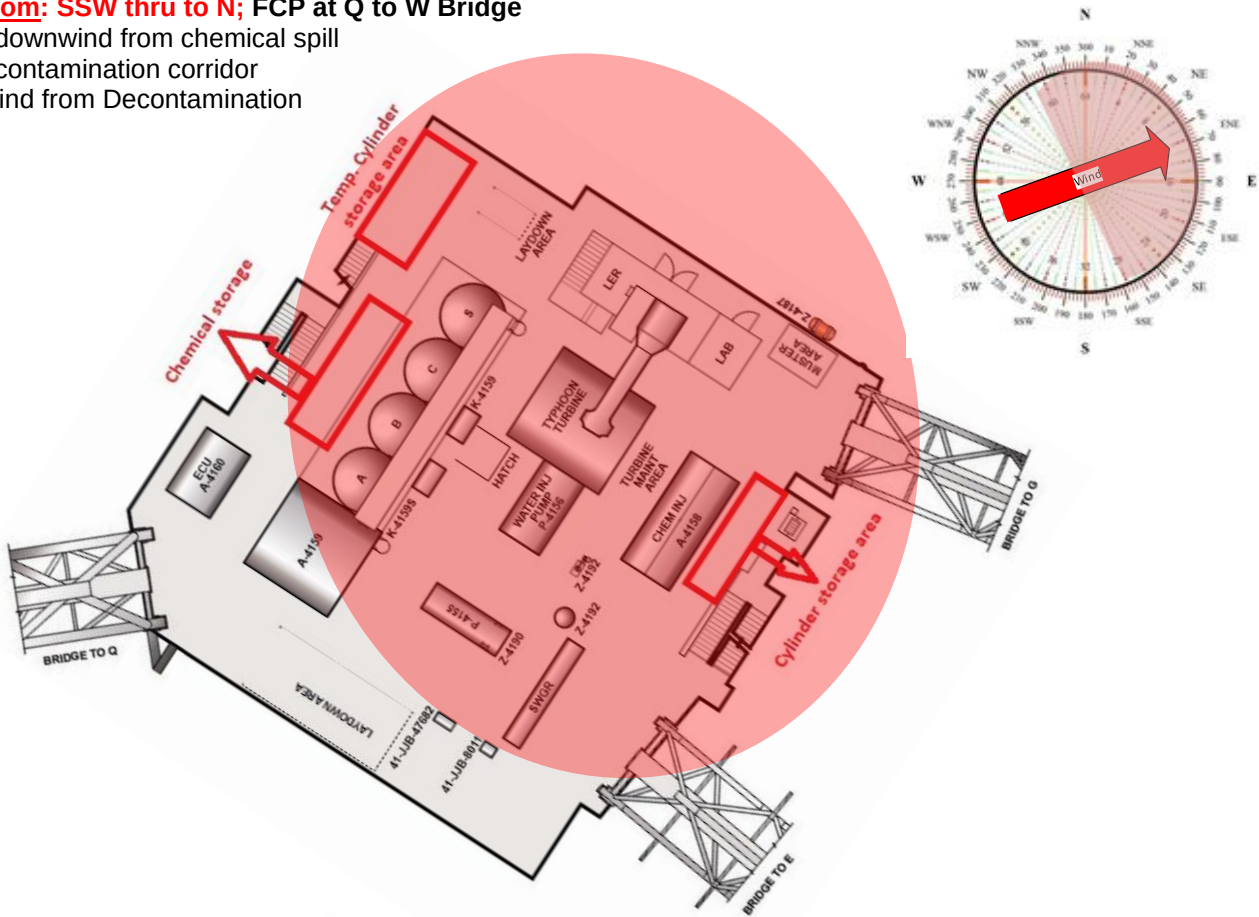
Personal Protective Equipment	A	B	C
SCBA			
Full Face Filter Mask with filter Cartridge			
Totally encapsulated chemical-protective suit with vapour barrier			
Hooded chemical resistant suit (one or two piece)			
Coveralls			
Outer Gloves, chemical resistant			
Inner Gloves, chemical resistant			
Steel Toe capped boots, chemical resistant			
Hard hat (worn under suit)			
Hard hat			
Face Shield			
Other PPE as required by site risk assessment prior entry			
	Requirement		
	Optional		
	Inside Spill Kit		

Wind direction from: SSW thru to N; FCP at Q to W Bridge

Hot zone – Area downwind from chemical spill

Warm zone – Decontamination corridor

Cold zone – Upwind from Decontamination



Coagulant: T-4158A (Water Injection)

Non-combustible

Hazard ID: Harmful if swallowed / Causes skin irritation / Causes serious eye damage / Harmful to aquatic life

First Aid: Allow the victim to rest in a well ventilated area / Skin, place the victim under a deluge shower for at least 15 mins / Eyes: Immediately flush eyes with running water for at least 15 mins.

Firefighting: Dry chemical, foam, carbon dioxide, water fog.

Polyelectrolyte: T-4158B (Water Injection)

Non-combustible

Hazard ID: Nuisance dusts may cause irritation / Dust may cause burning, itching resulting in reddening of the eyes & skin

First Aid: If inhaled, move to source of fresh air / Eyes - Immediately flush for 15 mins, Skin, clean with plenty soap / water.

Firefighting: Water spray or fog / Co2 / Dry Power Extinguishers

Biocide-G: T-4158B (Water Injection)

Non-combustible

Hazard ID: Causes skin irritation, wash thoroughly after handling. Vapours may cause eye irritation.

First Aid: Skin – wash thoroughly with soap and water, Eye – immediately flush with water for 15 mins.

Firefighting: Any means suitable for surrounding fire

Scale Inhibitor: T-4158D (Water Injection)

Non Combustible

Hazard ID: Corrosive to the skin. Causes burns / Corrosive to eyes. Causes burns. / Toxic if swallowed. May cause burns to mouth, throat and stomach.

First Aid: Eyes - Immediately flush the eye(s) continuously with lukewarm, gently flowing water for at least 20-60 minutes / Wash affected area with soap and mild detergent for at least 20 - 60 minutes /

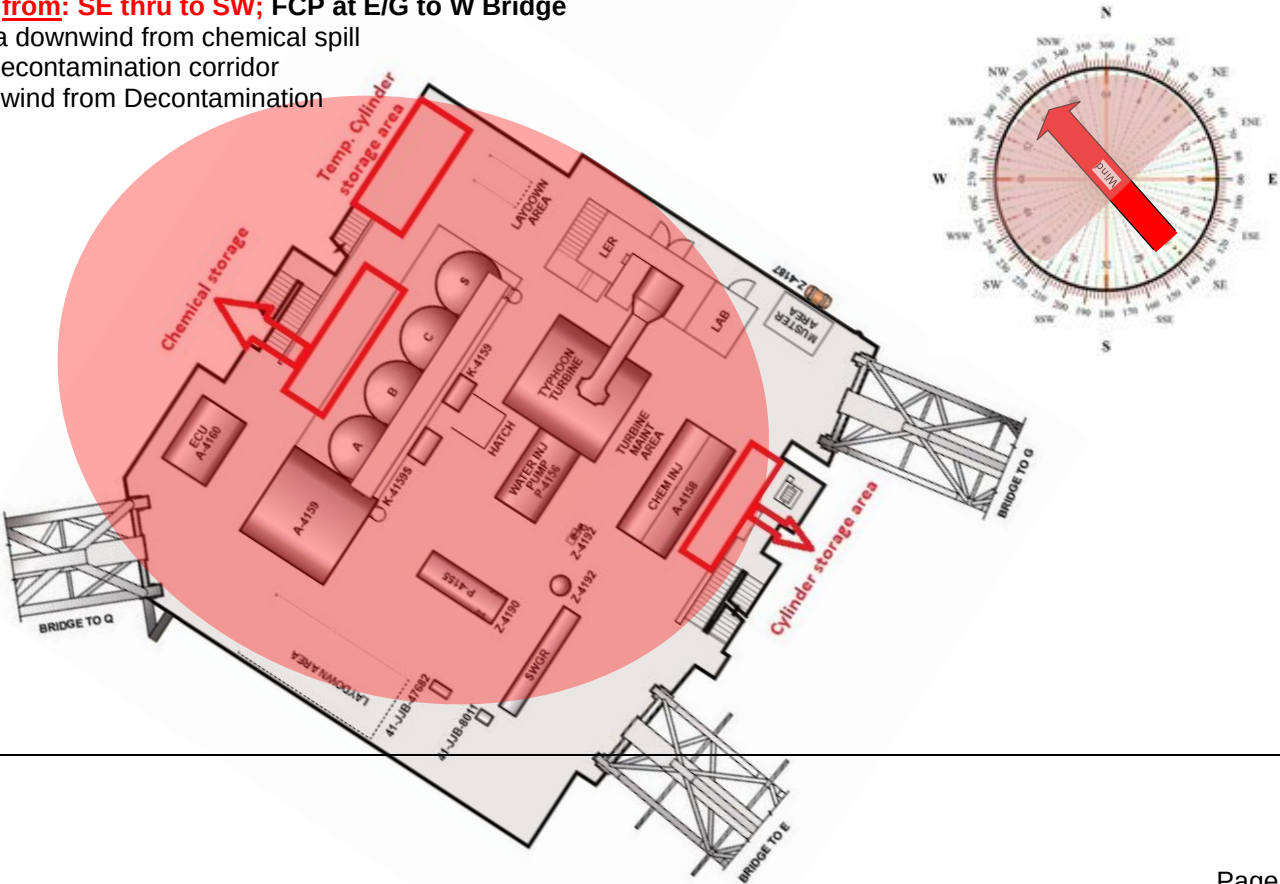
Firefighting: Dry chemical, foam, carbon dioxide, water fog.

Wind direction from: SE thru to SW; FCP at E/G to W Bridge

Hot zone – Area downwind from chemical spill

Warm zone – Decontamination corridor

Cold zone – Upwind from Decontamination



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4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/ External Document	Document Referencing
Offshore Pre-Incident Procedures	IS-FOP-PRC-450	Internal	Upward
Emergency Management Procedure	IS-HSE-PRC-001	Internal	Sideward
PS-1 Emergency Procedure	IS-FOP-PRC-210	Internal	Sideward
Offshore Tactical Response Procedure	IS-HSE-PRC-003	Internal	Sideward

5.0 DEFINITIONS

TABLE 2: KEY TERMS DEFINED

Term	Definition
NA	NA

APPENDIX

APPENDIX A - REVISION HISTORY LOG

Revision Number: 00

Date: 17/07/2023

Item Revised	Revision Description	Page No.
-	New Work Instruction	-
Remarks: Replaces TRP 011 of IS Offshore Tactical Response Procedure (IS-HSE-PRC-003)		

APPENDIX B - LIST OF ABBREVIATIONS

Term	Definition
CCR	Central Control Room
ERT	Emergency Response Team
IC	Incident Commander
PIP	Pre-Incident Plans
PPE	Personal protective equipment
OIS	Offshore Installation Superintendent
TR	Temporary Refuge